

Design, Fabrication and Analysis of MZI Devices in SOI Photonics

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Abstract

This report presents the design of a silicon-based unbalanced Mach–Zehnder interferometer (MZI) implemented on a silicon-on-insulator (SOI) platform using strip waveguides. Numerical simulations and analytical modeling are employed to justify the design parameters. Experimental measurements are compared with simulation results to validate the overall design and fabrication process. The extracted group index is **4.04** with a measured FSR of **2.82 nm**, in good agreement with simulation ($n_g \approx 4.07$).

1. Introduction

Integrated photonics has significantly advanced optical communication systems by enabling high-speed, compact, and low-power devices [1], [2]. Among these, the Mach–Zehnder interferometer (MZI) is a key component widely used in applications such as optical modulation, switching, and routing due to its sensitivity to phase differences between two optical paths [1], [4]. In silicon photonics, MZIs are typically implemented on silicon-on-insulator (SOI) platforms using strip waveguides. In this work, the waveguide properties were first analyzed using Lumerical, considering both TE and TM modes, and a corner analysis was performed to estimate the variation in group index. Based on these results, several MZI structures with different path length differences were designed using KLayout and simulated to obtain their transmission response. The designs were then prepared for fabrication, and the experimental results were compared with simulations to evaluate the performance and the impact of fabrication variations.

2. Theory

2.1 Mach–Zehnder Interferometer

The Mach–Zehnder interferometer (MZI) consists of a splitter and a combiner connected by two optical paths. The input optical field is divided equally into the two arms, propagates through waveguides, and is recombined at the output. The output intensity depends on the phase difference between the two arms. In an unbalanced MZI, a path length difference ΔL is introduced between the two arms. This creates a phase difference that varies with wavelength, resulting in constructive and destructive interference at the output. Neglecting propagation loss and assuming identical waveguides, the output intensity can be expressed as:

$$I_o = \frac{1}{2} I_i (1 + \cos(\beta \Delta L))$$

where $\beta = \frac{2\pi}{\lambda} n_{\text{eff}}$ is the propagation constant, n_{eff} is the effective index, and λ is the wavelength.

2.2 Free Spectral Range (FSR)

The periodicity of the interference pattern is characterized by the free spectral range (FSR), defined as the wavelength spacing between consecutive transmission peaks:

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L n_g(\lambda)}$$

where ΔL is the path length difference and n_g is the group index of the waveguide.

2.3 Group Index

The group index describes the propagation of optical pulses and accounts for dispersion in the waveguide. It is defined as:

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}(\lambda)}{d\lambda}$$

This parameter plays a critical role in determining the FSR and overall spectral response of the MZI.

3. Modeling and Simulation

Corner Analysis (Waveguide Group Index)

A corner analysis was performed to evaluate the sensitivity of the waveguide group index to fabrication variations in width and height. Five simulations were conducted at a wavelength of 1550 nm, considering both TE and TM polarization modes. The selected dimensions represent typical process variation ranges for silicon photonic waveguides. The resulting group index values (n_g) for each case are summarized in **Table I**. As shown in **Table I**, the TE mode exhibits relatively small variation compared to the TM mode, indicating higher stability with respect to dimensional changes. In contrast, the TM mode shows stronger dependence on waveguide geometry, particularly with respect to height variations.

TABLE I
GROUP INDEX (n_g) FOR TE AND TM MODES UNDER WAVEGUIDE DIMENSION VARIATIONS AT $\lambda = 1550$ nm

NO	Width(nm)	Height(nm)	TE group index	TM group index
1	500	220	4.177	3.735
2	470	215.3	4.225	3.475
3	470	223.1	4.241	3.724
4	510	215.3	4.156	3.573
5	510	223.1	4.170	3.813

The simulated optical mode profile of the waveguide cross-section is illustrated in **Fig. 1**, where strong field confinement within the silicon core can be observed. This confirms proper waveguiding behavior at the operating wavelength of 1550 nm. These results are essential for understanding the expected performance spread of interferometric devices such as Mach–Zehnder interferometers under fabrication tolerances.

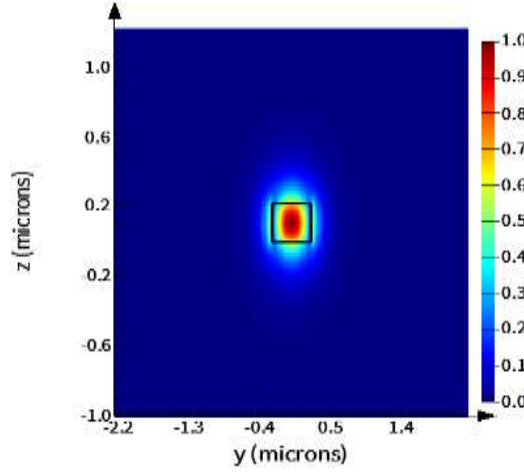


Fig. 1. Simulated quasi-TE mode profile showing field confinement in the waveguide(500 nm×220 nm) at 1550 nm.

4. Layout

The photonic circuit layout was designed using KLayout, as shown in **Fig. 2**. The design consists of five Mach–Zehnder interferometers (MZIs) with different path length differences (ΔL), all operating in TE polarization at 1550 nm. Each device includes grating couplers with a 127 μm pitch for fiber array compatibility. The waveguides were designed with a bend radius of 5 μm to minimize loss while maintaining compactness and compliance with fabrication design rules.

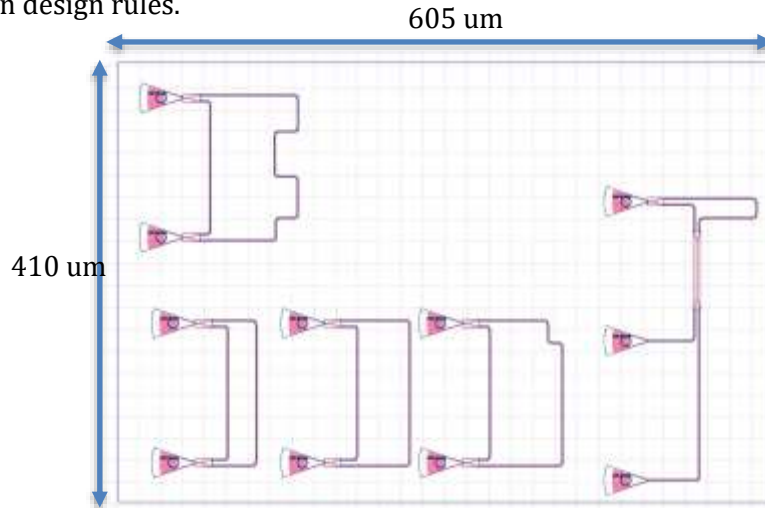


Fig. 2. KLayout draft of five TE Mach–Zehnder interferometers at 1550 nm.

5. Circuit simulation

The designed layout was translated into a circuit-level model and simulated using Lumerical INTERCONNECT. The simulation was performed using compact models from the EBeam process design kit (PDK), enabling accurate representation of waveguide components, couplers, and grating interfaces. An Optical Network Analyzer (ONA) was used to evaluate the transmission response of each MZI over a wavelength range centered at 1550 nm. The simulated spectra were used to extract key parameters such as the free spectral range (FSR) and group index (ng), which were later compared with experimental measurements.

6. Fabrication

The devices were fabricated on a silicon-on-insulator (SOI) wafer using electron beam lithography (EBL), following the process described in the course . The SOI wafer consists of a ~ 220 nm silicon device layer on top of a buried oxide (BOX), which acts as the lower cladding. First, a negative electron-beam resist (HSQ) was spin-coated onto the silicon surface and baked to form a thin, uniform layer. The sample was then exposed using EBL, where the designed photonic patterns were written directly into the resist. After exposure, the resist was developed in a tetramethylammonium hydroxide (TMAH) solution, leaving only the exposed regions. The pattern was subsequently transferred into the silicon layer באמצעות inductively coupled plasma (ICP) etching, resulting in fully etched waveguide structures. Following etching, the sample was cleaned and inspected. A silicon dioxide layer (~ 2 μm) was then deposited over the chip to provide upper cladding and environmental protection. Finally, the wafer was diced into smaller chips for testing.

6. Experimental Data

The experimental data corresponding to the MZI with the smallest discrepancy relative to the simulation results was selected for further analysis. As shown in **Fig. 3**, the raw transmission spectrum of the MZIAA-517 device (TE mode) is presented. A baseline correction was then applied using a third-order polynomial fit to remove the grating coupler response.

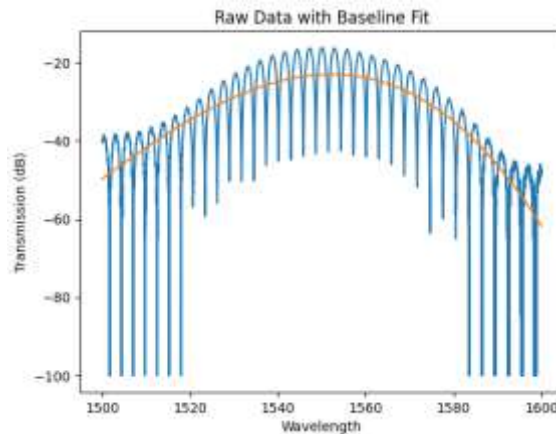


Fig. 3. Measurement and baseline correction of the MZIAA-517 (TE) device

The baseline-corrected transmission spectrum of the selected MZI is shown in **Fig. 4**. After removing the grating coupler envelope using a third-order polynomial fit, clear and periodic interference fringes are observed across the wavelength range. The free spectral range (FSR) was extracted directly from the corrected spectrum by measuring the spacing between adjacent interference minima and was found to be **2.82 nm**. Using this value, the group index of the waveguide was calculated to be **4.04**.

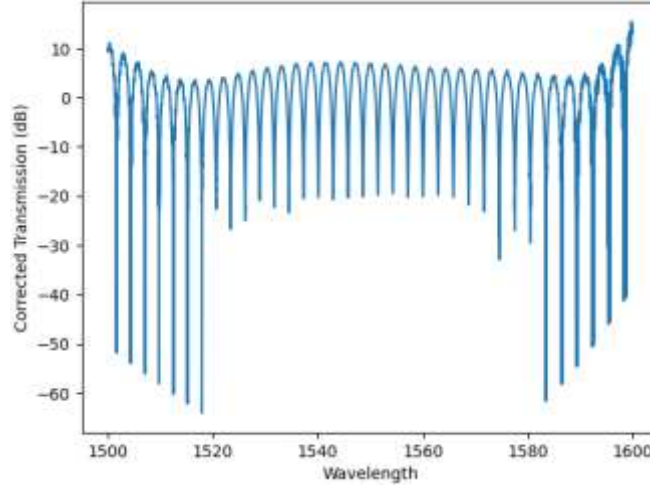


Fig. 4. Baseline-corrected transmission spectrum of the MZIAA-517 (TE) device

7. Measurement vs Simulation

The simulation results were extracted from the transmission spectrum of the selected MZI. Around 1550 nm, the simulated free spectral range (FSR) was found to be approximately **2.80 nm**, corresponding to a group index of **4.07**. These values are in very good agreement with the experimental results, where the FSR was measured as **2.82 nm** and the group index was calculated as **4.04**. The discrepancy in group index is about **0.7%**, indicating close agreement between simulation and measurement.

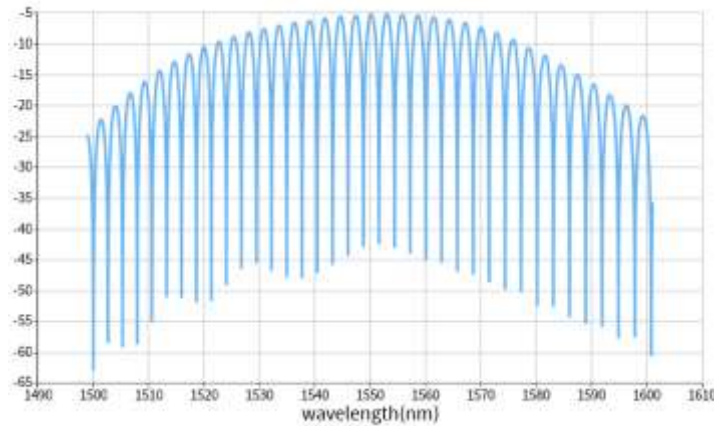


Fig. 5. Simulated transmission spectrum of the MZI at TE polarization near 1550 nm,

8. Conclusion Acknowledgments:

The MZI device was successfully designed, simulated using Lumerical INTERCONNECT, and experimentally characterized. After baseline correction, clear interference fringes were observed, enabling extraction of the free spectral range (FSR). The measured FSR of **2.82 nm** corresponds to a group index of **4.04**, which is in close agreement with the simulated values (FSR $\approx$ **2.80 nm**, $n_g \approx$ **4.07**), with a discrepancy of approximately **0.7%**. The experimental results also fall within the range predicted by the corner analysis.

Overall, the strong agreement between simulation and measurement validates the design and fabrication approach.

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